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1 最短路徑

1.1 basic

```
1 // c++ code
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 int main() {
6     // test comment
7     cout << "test string\n";
8 }
```

2 圖論

2.1 凸包問題

```
1 struct node{
2     int x,y;
3     bool operator<(const node &other) const{
4         return (x<other.x) || (x==other.x &&
5             y<other.y);
6     };
7 };
8 struct ConvexHull{
9     vector<node> vec;
10    int n;
11    void init(vector<node> &v){
12        vec=v;
13        n=v.size();
14    }
15
16    int cross(node a,node o,node b){
17        return
18            (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
19    }
20
21    vector<int> point;
22    vector<int> area;
23
24    void build(){
25        point=vector<int>(n+1);
26        area=vector<int>(n+1);
27        vector<node> up,down;
28        int upsum=0;
29        int downsum=0;
30        for(int i=0;i<n;i++){
31            while(up.size()>=2 &&
32                cross(up[up.size()-2],up[up.size()-1],vec[i])<=0){
33                upsum-=cross(up[up.size()-2],{0,0},up[up.size()-1]);
34                up.pop_back();
35            }
36            up.push_back(vec[i]);
37            if(up.size()>1)
38                upsum+=cross(up[up.size()-2],{0,0},up[up.size()-1]);
39
40            while(down.size()>=2 &&
41                cross(down[down.size()-2],down[down.size()-1],vec[i])>=0){
42                downsum-=cross(down[down.size()-2],{0,0},down[down.size()-1]);
43                down.pop_back();
44            }
45            down.push_back(vec[i]);
46            if(down.size()>1)
47                downsum+=cross(down[down.size()-2],{0,0},down[down.size()-1]);
48
49            area[i+1]=abs(upsum-downsum);
50
51            if(i+1==1) point[i+1]=1;
52            else point[i+1]=up.size()+down.size()-2;
53        }
54    }
55
56    double getarea(int p){
57        return area[p]/2.0;
58    }
59
60    int getnum(int p){
61        return point[p];
62    }
63 };
```

3 數學

3.1 公式

• 中文測試

$$\cdot \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$